Lab 2 – Smart City Product Specification

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) is a technical document intended for the **developers** of the Smart City project. It serves as a definitive technical reference for system implementation, outlining the functional and non-functional requirements necessary to build the initial prototype

1.2 Scope

The Smart City system is a data-driven urban navigation platform designed to mitigate the challenges of rapid urbanization and traffic congestion.

- The system will: provide a mobile/web interface, perform real-time traffic monitoring via crowdsourcing, generate AI-based alternative route and destination suggestions, and provide heat map visualizations.
- The system will not: include physical traffic light control, law enforcement database integration, autonomous vehicle communication systems, or full-scale smart-city physical infrastructure.

1.3 Definitions, Acronyms, and Abbreviations
MFCD: Major Functional Component Diagram.

Privacy-by-Design: An engineering approach where privacy is integrated into the system's architecture by default.
RWP: Real World Product.

Internet of Things (IoT): systems that combine sensing, computing, communication, and actuation to connect physical/cyber objects

Application Programming Interface (API): a set of features and rules that lets software interact with other software (a "contract" between programs)

Crowdsourced Reporting: gathering information by soliciting contributions from a large group of people, typically via the internet

Privacy-by-Design ("Privacy-First Design"): building systems so that, by design and by default, only the personal data necessary for a purpose are processed

Open Data (Open Data Access): data anyone can freely use, re-use, and redistribute, with at most attribution/share-alike requirements

Predictive AI / AI-Powered Forecasting: using data, statistical algorithms, and ML to estimate the likelihood of future outcomes

Anomaly Detection: identifying data patterns that deviate from expected behavior

Access Control: controls that enforce authorized access to information and system resources in line with policy

Citizen Feedback Loop: a two-way process where institutions act on and report back about citizen input to sustain engagement and trust

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1.5 Overview

This Software Requirements Specification (SRS) provides a technical roadmap for the development and implementation of the **Smart City prototype**. The remainder of the document is organized into **Section 2: Overall Description**, which defines the high-level technical foundations of the project. Specifically, this includes the **Product Perspective** (2.1) detailing the system architecture and external interfaces; **Product Functions** (2.2) outlining core features and their implementation status; **User Characteristics** (2.3) defining the roles and expertise of residents and city officials; and the technical **Constraints** (2.4) and **Assumptions and Dependencies** (2.5) that govern the system's development

2 Overall Description

2.1 Product Perspective

```
[ PRESENTATION LAYER ] <------> [ APPLICATION LAYER ]
(React Native / JS)             (Java Backend)
      |                               |
      |                               |
 +----------+                   +-------------+
 | Residents|                   |  AI Logic   |
 |   App    |                   |     &       |
 +----------+                   | Data Handling|
      |                         +-------------+
 +----------+                         |
 |  City    |                   +-------------+
 | Officials| <---------------> | External APIs|
 | Dashboard|                   | (Google Maps)|
```

Figure 1: Smart City Major Functional Component Diagram

This architectural design enables the system to operate as a cohesive urban management tool by facilitating data exchange across all tiers. The **Presentation Layer** serves as the primary gateway, providing mobile interfaces for residents to contribute crowdsourced traffic data and a web-based dashboard for city officials to access analytics. These interfaces communicate with the **Application Layer**, where the **Java-based backend** processes incoming data and integrates it with external **Google Maps, Google Places, and Heat Map APIs** to generate AI-driven route recommendations. All system data, including user profiles and historical congestion patterns, is persistently managed within the **Data Layer** using **Firebase** to ensure scalable storage and efficient retrieval for long-term urban planning

2.2 **Product Functions**

The following table describes the core features of the Smart City platform and their implementation status within the functional prototype.

Table 1: Smart City Feature Description and Prototype Implementation

Feature	Description	Prototype Implementation
Real-time Traffic Monitoring	Collects live traffic data via crowdsourced user inputs to provide real-time updates.	**Fully Implemented**
AI Route Recommendations	Uses predictive algorithms to suggest optimized travel paths based on live conditions.	**Fully Implemented**
Alternative Destinations	Suggests nearby locations to users to prevent overcrowding in specific urban areas.	**Partially Implemented**
Heat Map Visualization	Provides visual data representation of congestion levels for easier interpretation by users.	**Fully Implemented**
City Official Dashboard	An analytical interface for city planners to review and analyze urban traffic patterns.	**Fully Implemented**
Physical Traffic Control	Direct integration and control over physical city infrastructure like traffic lights.	**Eliminated**
Law Enforcement Integration	Direct data sharing and connectivity with police and law enforcement systems.	**Eliminated**
Autonomous Vehicle Systems	Communication protocols and systems for autonomous vehicle integration.	**Eliminated**

2.3 **User Characteristics**

Residents and Commuters: Primary users who require optimized navigation to reduce travel time. Expected to have a novice to intermediate level of smartphone proficiency.

City Officials and Planners: Users who utilize the data dashboard to make informed urban development decisions. Expected to have a high level of technical expertise in data analysis.

2.4 Constraints

Technical: The system is dependent on persistent high-speed internet connectivity for the real-time processing of crowdsourced data.

Legal/Ethical: The system must adhere to strict Privacy-by-Design standards to ensure that all user-contributed data is anonymized and securely stored.

2.5 Assumptions and Dependencies

Frameworks: Successful deployment depends on the continued stability and support of the React Native mobile framework and the Java development environment.

Third-Party Services: The navigation and visualization features assume the operational availability of Google Maps APIs and Firebase cloud services.